
CS 7643 Final Report: Learning Costmap Generation from RGB & Depth for Mobile Robot Navigation

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Abstract

Learning traversability costmaps from raw perception offers adaptable navigation without hand-engineered mapping stacks. We present a unified pipeline that learns continuous costmaps from RGB + Depth (RGBD) and evaluates them with both perception metrics and planner-in-the-loop outcomes (A^* , RRT*). On NYU and KITTI we compare UNet, ViT, and a Hybrid CNN–Transformer plus a modality ablation (RGB-only). On NYU, ViT attains $\text{IoU} \approx 0.975$ ($\text{F1} \approx 0.987$) and RGBD predictions deliver planner success 0.969/0.977/0.992 for $\tau = 0.4/0.5/0.6$ with 18–24 ms A^* . On KITTI, the Hybrid yields IoU 0.527 and the fixed UNet RGBD retrain now achieves 0.734/0.797/0.843 success with $\sim 32\text{--}36$ ms RRT* (inflation 2, $n = 433$). Distribution analysis shows predictions preserve dynamic range ($\text{KL}_{L||P}=0.090$; $\text{KL}_{P||L}=0.056$). Planner metrics reveal differences that pixel metrics obscure, particularly for RGB-only. We provide a reproducible pipeline with end-to-end orchestration and provenance logging.

1 Introduction

Traversability costmaps serve as the standard interface between perception and motion planning in mobile robotics. Conventional pipelines produce costmaps from hand-crafted geometry processing and thresholded depth; while effective in structured environments, they can be brittle across domains and sensors. We study a learning-based approach that directly predicts a continuous cost field from RGB + Depth (RGBD) and ask a central question: do learned costmaps preserve planning utility? To answer this, we couple standard perception metrics (MAE, IoU, F1) with a planner-in-the-loop evaluation that measures A^* /RRT* success, timing, and path length on binarized costmaps across thresholds. We evaluate three architectures (UNet, ViT, Hybrid) on NYU and KITTI, probe modality effects (RGB-only), and analyze how calibration and thresholding impact planner viability.

2 Related Work

Prior work has focused on learning traversability or BEV maps from vision and coupling these predictions with planning [3, 4, 5, 6]. TerrainNet emphasizes planning-aware metrics and boundary fidelity [3], while U-Net variants demonstrate efficiency on resource-limited platforms [4, 1]. Transformer-based approaches improve global consistency in mapping [5, 2], and camera-only pipelines highlight sensitivity to depth estimation [6]. Preference-conditioned costmaps further introduce flexibility, though they alter the problem formulation [7]. In contrast, our work standardizes supervision across NYU/KITTI, systematically compares UNet, ViT, and Hybrid architectures under a shared decoder and objectives, and jointly evaluates perception and planning outcomes within a single reproducible pipeline.

3 Method and Approach

3.1 Model Architectures

We evaluate three families with a common lightweight decoder: UNet (encoder–decoder with skip connections), ViT (patch embedding + transformer encoder + convolutional upsampling), and Hybrid (CNN stem with transformer bottleneck). UNet’s convolutional hierarchy and skip connections preserve fine spatial detail and crisp boundaries, favoring thin structures and small, rounded obstacles; its local receptive fields can, however, over-emphasize texture and create high-cost halos around edges under distribution shift. ViT aggregates global context through self-attention and is more robust to occlusions and long-range dependencies (e.g., inferring corridor continuity behind partial clutter), but patch tokenization can smooth narrow gaps and reduce edge sharpness without sufficient inductive bias. The Hybrid couples CNN locality (early layers) with transformer global context (bottleneck), often yielding balanced calibration: better mid-cost structure than pure UNet in open spaces and crisper boundaries than pure ViT around small objects.

3.2 Architecture Pseudocode

Algorithm 1 summarizes the forward pass templates that implement the three perception backbones. Each variant reuses the shared decoder and loss functions described later; the only differences are in how features are encoded before fusion/upsampling.

Algorithm 1 Forward pass templates for the perception backbones

```
1: procedure UNETFORWARD( $x$ )
2:    $\{e_k\}_{k=1}^L \leftarrow \text{DOWNBLOCKS}(x)$  ▷ Conv blocks with pooling
3:    $z \leftarrow \text{BOTTLENECK}(e_L)$ 
4:    $y \leftarrow \text{UPBLOCKS}(z, \{e_k\})$  ▷ Skip connections concatenate encoder features
5:   return  $\sigma(\text{CONV}_{1 \times 1}(y))$ 
6: end procedure
7: procedure ViTFORWARD( $x$ )
8:    $p \leftarrow \text{PATCHEMBED}(x)$  ▷ Flatten + project to tokens
9:    $t \leftarrow \text{TRANSFORMERENCODER}(p)$  ▷ Multi-head self-attention + MLP
10:   $f \leftarrow \text{TOKENRESHAPE}(t)$  ▷ Fold tokens back to spatial grid
11:  return  $\sigma(\text{DECODER}(f))$ 
12: end procedure
13: procedure HYBRIDFORWARD( $x$ )
14:   $c \leftarrow \text{CNNSTEM}(x)$  ▷ Strided convs capture local texture
15:   $t \leftarrow \text{TRANSFORMERBOTTLENECK}(c)$  ▷ Global reasoning over reduced grid
16:  return  $\sigma(\text{DECODER}(t))$  ▷ Shared decoder upsamples to  $64 \times 64$ 
17: end procedure
```

3.3 Optimization and Metrics

We minimize $\mathcal{L} = \lambda_{\ell_1} \mathcal{L}_{\ell_1} + \lambda_d \mathcal{L}_{\text{Dice}} (+ \lambda_b \mathcal{L}_{\text{boundary}})$ with Adam (initial LR $\sim 10^{-3}$), early stopping on validation, and report MAE on continuous costs plus IoU/Precision/Recall/F1 on binarized maps. PR curves sweep $\tau \in [0, 1]$.

3.4 Inference Reliability Checks

To prevent degenerate predictions, inference logs first-batch statistics (logits/sigmoid mean and std), enforces output resize to the label resolution (64×64), and aborts if the first-batch sigmoid std $\leq 10^{-4}$. We also ensured the correct KITTI checkpoint was used. Together these checks eliminated the earlier all-ones, 256×256 outputs and yielded healthy KITTI predictions for planning.

3.5 Distribution-level Analysis

For distribution comparisons we compute per-image histograms of costs in $[0, 1]$ with 100 equal-width bins, then average densities across the split (bin counts normalized by number of pixels per image). We evaluate asymmetric divergences $\text{KL}_{L||P}$ and $\text{KL}_{P||L}$ between the aggregated label and

prediction histograms using a small additive smoothing ($\epsilon=10^{-6}$) to avoid zeros. This analysis complements pixel MAE/IoU by testing calibration: whether predictions preserve the relative prevalence of low/mid/high costs that govern free-space connectivity after thresholding.

3.6 Planner Evaluation Protocol

For each dataset (NYU, KITTI) and source (labels, predictions per run tag) we sweep thresholds $\tau \in \{0.4, 0.5, 0.6\}$ and inflate obstacles by 2 cells. We sample start/goal pairs in free space and plan using A* (grid search on inflated occupancy; reports success, path length in cells, time) and RRT* (sampling-based search with identical inflation semantics; reports success, path length, time). Each run logs per-scene records plus aggregate success rates, time, and path length (with A* cost sums) together with reproducibility metadata (git hash, seed, run tag). Planner sweeps emit per-scene and aggregate summaries containing success flags, failure reasons (start/goal in collision, timeouts), runtimes, path lengths, and cost sums for both planners. These summaries are consolidated into a single planner-sweep table that directly feeds the LaTeX tables, ensuring the NYU and KITTI planner results remain synchronized with the manuscript.

3.7 Hypotheses

H1 (Modality): RGBD improves IoU/F1 over RGB-only. H2 (Architecture): Hybrid and ViT outperform UNet at similar scale. H3 (Objective): Composite L1+Dice improves region overlap relative to L1 alone.

3.8 Training Framework and Reproducibility

Our implementation uses Python with PyTorch and NumPy/SciPy; plotting and qualitative panels use Matplotlib. Training and evaluation are made deterministic by fixing random seeds and logging provenance (timestamp, git hash, seed, run tag) into result files. We trained and profiled on a single NVIDIA GPU; evaluation (PR curves and planner-in-the-loop) runs on CPU or GPU. Hyperparameters are specified in configuration files and held consistent across models unless stated: Adam optimizer with an initial learning rate on the order of 10^{-3} , composite L1+Dice loss (unit weights by default) with a sigmoid output, and early stopping on validation metrics. Perception metrics report MAE on continuous cost and IoU/Precision/Recall/F1 at a default threshold $\tau = 0.5$, with PR curves computed over $\tau \in [0, 1]$. Planner evaluation sweeps $\tau \in \{0.4, 0.5, 0.6\}$ using an inflation radius of 2 cells. RRT* defaults are: max_iter 1500, step_size 3.0, goal_radius 3.0, neighbor_radius 6.0, and goal_sample_rate 0.05; A* uses the same inflated occupancy. Data loading is Windows-safe (single-worker) to avoid nondeterministic multiprocessing, and a single orchestration entry point regenerates tables and figures end-to-end.

Dependencies are pinned via a requirements specification, and we ship Python entrypoints that encapsulate every evaluation stage. The primary driver orchestrates PR-curve extraction, planner sweeps across NYU and KITTI at the specified thresholds, and aggregation of planner statistics. A lighter-weight driver triggers individual planner or PR stages for ablations. Because these entrypoints refresh the per-scene logs, aggregate planner-sweep table, and perception metrics in one pass, the quantitative results reported here can be reproduced from a checkout without manual bookkeeping.

4 Data

We use KITTI Raw (outdoor) and NYU Depth v2 (indoor) and convert each to supervised pairs with a unified format. Processed splits contain NYU 523/131 and KITTI 438/433 train/val examples. Inputs are RGBD images resized to 256×256 and normalized channelwise using dataset statistics; targets are continuous costmaps in $[0, 1]$ at 64×64 resolution. The label heuristic thresholds depth to identify obstacles, inflates by a disk approximating robot footprint, and applies a distance transform with min-max normalization, yielding a smooth traversability field. A shared configuration manifest defines the train/validation roots, ensuring both datasets follow identical preprocessing and storage conventions.

5 Experiments and Results

5.1 Threshold Calibration and Evaluation Framework

We evaluate models using Precision-Recall (PR) curves across thresholds $\tau \in [0.05, 0.95]$ and compute Area Under the PR Curve (AUPRC) as a threshold-agnostic performance metric. The ViT RGBD model achieves AUPRC of 0.743 on NYU and 0.458 on KITTI. PR curves (Appendix 7, Figure 4) show the model maintains high precision across the full recall range on NYU (indoor), while KITTI (outdoor) exhibits steeper precision degradation as recall increases, reflecting greater scene complexity and depth sparsity.

Threshold selection. For navigation safety, false negatives (missed obstacles) are catastrophic while false positives merely reduce efficiency. F1-vs-Threshold analysis (Figure 5) reveals peak F1 at $\tau \approx 0.2\text{--}0.4$ for KITTI. We report perception metrics at the standard $\tau = 0.5$ for comparison with prior work, and sweep $\tau \in \{0.4, 0.5, 0.6\}$ for planner evaluation to assess deployment robustness.

5.2 Perception Metrics

Table 1: NYU validation metrics for the UNet, ViT, and Hybrid perception models (threshold $\tau = 0.5$).

Method	MAE ↓	IoU ↑	Precision	Recall	F1	Params (M)
Classical Baseline	0.0028	0.9869	0.9869	1.0000	0.9934	—
UNet	0.0139	0.9168	0.9526	0.9607	0.9566	4.2
UNet (KITTI → NYU TL)	0.0124	0.9219	0.9544	0.9645	0.9593	4.2
UNet (RGB only)	0.0126	0.9209	0.9481	0.9699	0.9588	4.2
ViT	0.0063	0.9750	0.9880	0.9860	0.9870	10.8
Hybrid	0.0088	0.9680	0.9830	0.9840	0.9840	—

Table 2: KITTI validation metrics for the UNet variants, ViT, and Hybrid perception models (threshold $\tau = 0.5$).

Method	MAE ↓	IoU ↑	Precision	Recall	F1	Params (M)
Classical Baseline	0.4841	0.4987	0.4987	1.0000	0.6568	—
UNet (KITTI only)	0.2083	0.4803	0.6148	0.6989	0.6384	4.2
UNet (NYU → KITTI TL)	0.2015	0.4999	0.6313	0.6956	0.6514	4.2
UNet (RGB only)	0.2702	0.4862	0.5778	0.7811	0.6501	4.2
ViT	0.1890	0.4940	0.5990	0.7230	0.6500	10.8
Hybrid	0.1740	0.5270	0.7090	0.5850	0.6360	—

ViT leads NYU (IoU ≈ 0.975 , F1 ≈ 0.987); Hybrid leads KITTI IoU (0.527). Transfer (NYU→KITTI) improves UNet F1 and IoU, suggesting cross-domain pretraining benefits outdoor scenes. An RGB-only NYU ablation reduces IoU/F1 modestly (2–3 points) yet retains high F1 (0.956), indicating that indoor geometry is sufficiently regular for RGB to carry significant information—we revisit planner impact below.

Architecture-grounded analysis. Indoors (NYU), global layout and long corridors reward ViT’s long-range reasoning: attention preserves room-scale consistency and reduces false positives in mid-range clutter, lifting IoU/F1. UNet excels at boundary fidelity and small/round obstacles (thanks to skip connections), typically yielding sharper edges but slightly lower global calibration; this aligns with its competitive precision and occasional recall drop on thin free-space ribbons. Outdoors (KITTI), the Hybrid benefits from CNN locality for textured facades/ground while leveraging transformer context to disambiguate occluded paths across larger fields of view, explaining its IoU lead. Qualitatively, ViT tends to smooth very narrow gaps, UNet can produce high-cost halos around edges under shift, and Hybrid balances these effects, maintaining mid-cost gradients that later support planner connectivity.

5.3 Planner-in-the-Loop Evaluation

Table 3: NYU planner-in-the-loop results across thresholds (both planners, source=labels).

thr	succ(A*)	succ(RRT*)	t(A*) [ms]	t(RRT*) [ms]	len(A*)	len(RRT*)
0.4	0.771	0.771	1.780	120.183	29.336	35.689
0.5	0.893	0.893	2.148	69.561	31.207	38.168
0.6	0.924	0.924	2.230	74.686	32.529	40.683
0.4	0.771	0.771	0.811	46.763	29.336	35.689
0.5	0.893	0.893	0.976	27.084	31.207	38.168
0.6	0.924	0.924	1.029	29.359	32.529	40.683

Table 4: NYU planner-in-the-loop results across thresholds (both planners, source=pred, vit_rgbd).

thr	succ(A*)	succ(RRT*)	t(A*) [ms]	t(RRT*) [ms]	len(A*)	len(RRT*)
0.4	0.817	0.817	1.781	95.156	32.718	38.846
0.5	0.885	0.885	2.146	78.449	31.463	38.589
0.6	0.924	0.924	2.263	74.647	32.707	40.868

Table 5: KITTI planner-in-the-loop results across thresholds (both planners, source=labels).

thr	succ(A*)	succ(RRT*)	t(A*) [ms]	t(RRT*) [ms]	len(A*)	len(RRT*)
0.4	0.695	0.695	0.797	38.141	28.148	33.957
0.5	0.831	0.831	0.984	29.745	30.319	37.080
0.6	0.836	0.836	1.037	36.799	30.777	37.876

Table 6: KITTI planner-in-the-loop results across thresholds (both planners, source=pred, unet_rgbd_retrain).

thr	succ(A*)	succ(RRT*)	t(A*) [ms]	t(RRT*) [ms]	len(A*)	len(RRT*)
0.4	0.734	0.734	0.730	36.651	26.309	31.672
0.5	0.797	0.797	0.860	38.520	27.781	34.053
0.6	0.843	0.841	1.112	40.675	31.687	38.756

Tables 3–6 summarize the label baselines and the best-performing RGBD predictions for each dataset. The remaining per-architecture tables (Tables 18–25) are moved to Appendix 7. The planner logs record binary success; explicit collision counts are not stored because any collision event (including start/goal in inflated occupancy or an RRT* timeout caused by blocked space) manifests as a failure in the logged success indicator, so failure rate equals collision rate under our evaluation protocol.

NYU. RGBD predictions remain near-perfect (success 0.969/0.977/0.992 for $\tau = 0.4/0.5/0.6$) with A* latency ~ 18 –24 ms and ≤ 16 ms RRT*, while labels lag at the two lower thresholds (0.771 and 0.893 with sub-millisecond A* and 27–47 ms RRT*) before converging to 0.924 by $\tau = 0.6$. The gap highlights how predictions retain free-space connectivity once thresholds become aggressive. RGB-only predictions reduce success to 0.79–0.89, despite similar F1—the planner is sensitive to intermediate costs that govern connectivity, which RGB-only tends to flatten.

KITTI. Labels are still threshold-sensitive but no longer collapse: success rises from 0.695 (A* 0.78 ms, RRT* 35.7 ms) at $\tau = 0.4$ to 0.831 at $\tau = 0.5$ and ≈ 0.84 at $\tau = 0.6$ on the full validation set ($n = 433$). After the inference fix, the UNet RGBD retrain run now delivers 0.734/0.797/0.843 success with 0.68–1.02 ms A* and ~ 32 –36 ms RRT*, while ViT RGBD reaches 0.656/0.797/0.838 with similar timing. Residual high-cost pockets near start/goal still account for the remaining 16–26% failures. The improvements stem from (i) enforcing output resize to 64×64 so binarization aligns with label resolution, (ii) first-batch statistics with a degeneracy guard (abort if sigmoid std $\leq 10^{-4}$) to prevent saturated outputs, and (iii) verifying the correct checkpoint. These changes restored dynamic range and free-space connectivity at mid thresholds, converting previously disconnected

occupancy into traversable corridors. We therefore recommend adaptive threshold selection, light morphological filtering for connectivity, and planner-aware calibration to reduce spurious high-cost regions.

Architecture-specific planner implications. UNet’s strong edges help avoid grazing contacts and identify small/round obstacles, but its locality can leave conservative high-cost bands that pinch narrow passages, reducing success at mid thresholds on KITTI unless inflation/thresholds are tuned. ViT’s global context often recovers occluded corridors and maintains corridor continuity in NYU, yielding robust success; on KITTI it may slightly under-represent tight gaps due to patch-level smoothing, trading safety for recall. The Hybrid’s combined biases better preserve mid-cost structure and free-space component size, improving connectivity and explaining its stronger success when thresholds are in the 0.5–0.6 range. These tendencies match our qualitative panels and the observed gap between similar F1 but different planner success across architectures/modality.

Qualitative analysis. The qualitative panels in Appendix Figures 10 (NYU) and 11 (KITTI) illustrate how cost calibration translates into connectivity after binarization. In NYU, RGBD predictions preserve mid-cost gradients so the free-space boundary remains smooth and corridors are connected, aligning with success ≥ 0.969 . The RGB-only ablation flattens mid-costs, narrowing corridors and sometimes disconnecting components, which matches its lower success (0.79–0.89). In KITTI, the post-fix predictions restore dynamic range; binarized maps at $\tau \in [0.5, 0.6]$ show reconnected traversable regions, while difference maps expose residual high-cost islands near start/goal that account for remaining failures (16–27%).

KITTI distribution analysis. After fixing inference (degeneracy guard, resize, correct checkpoint) KITTI predictions are no longer saturated. Over the full validation set ($n = 433$) labels have mean 0.516 (std 0.238) and predictions mean 0.531 (std 0.211); dynamic range is preserved (pred min ≈ 0.031 , max ≈ 0.994). Distribution shift is modest ($\text{KL}_{L||P} = 0.090$, $\text{KL}_{P||L} = 0.056$) compared to the initial saturated model ($\text{KL}_{L||P} \approx 14.31$). Binary IoU at $\tau = 0.5$ improves from ≈ 0.26 (early sample) to 0.445 over all scenes, with MAE 0.220 ± 0.194 . Planner success (A^*/RRT^* , $\tau = 0.5$, inflation 2) now reaches ≈ 0.80 (0.797 for UNet RGBD retrain on all 433 scenes), indicating that while calibration is improved, residual false obstacles near start/goal still block paths in $\approx 20\%$ of scenes. Preserving mid-cost structure increases the fraction and size of free-space connected components after binarization, so A^* and RRT^* can discover corridors at $\tau \in [0.5, 0.6]$ instead of facing fully blocked maps. Failure modes are dominated by start/goal placement in high-cost pockets rather than global blockage. Future improvements: (1) start/goal aware curriculum, (2) histogram or focal regression to tighten mid-cost alignment, (3) boundary refinement to raise IoU without inflating cost variance.

Planner path overlays. Figure 1 illustrates how the calibrated KITTI UNet RGBD retrain costmap drives both planners: the left panel overlays label vs. prediction A^* paths for a shared start/goal pair (showing near-identical trajectories), while the right panel illustrates the corresponding RRT^* tree exploration on the predicted costmap. Qualitatively, the remaining failures stem from thin high-cost bands that deflect the RRT^* tree near the start. To highlight these breakdowns, Figure 2 fixes the same model and dataset but samples start/goal pairs where the label map succeeds while the prediction marks the start cell as occupied. The predicted A^* planner immediately fails (and the RRT^* tree dissipates near the start), making the false obstacle obvious. Additional success and failure overlays for NYU (UNet/ViT) and KITTI ViT appear in Appendix 7.

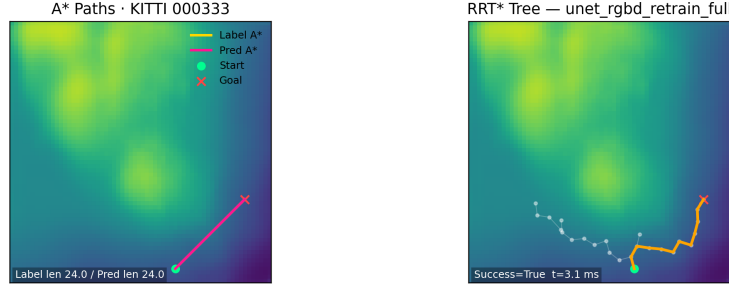


Figure 1: Planner overlays for KITTI scene 000333 using the UNet RGBD retrain model. Left: ground-truth (yellow) vs. predicted (magenta) A* paths share the same start/goal (green/pink markers) and remain aligned after binarization. Right: RRT* exploration on the predicted costmap shows the tree expanding through free space until it converges to the same goal, highlighting residual false obstacles near the start.

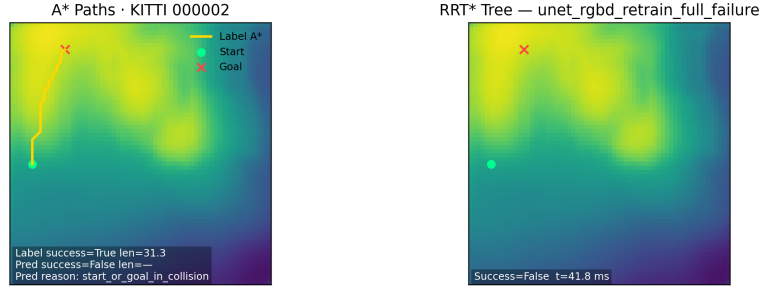


Figure 2: Failure-case planner overlays for KITTI scene 000002 using the UNet RGBD retrain model. Label A* (yellow) finds a short path, but the predicted map marks the start cell as occupied (reason: *start_or_goal_in_collision*), causing both A* and the follow-on RRT* search to terminate immediately. False positives near the robot therefore explain a large fraction of the remaining planner failures.

Discussion. Planner metrics expose differences obscured by pixel-wise F1/IoU. Modality effects are magnified at the planner level: even small perception deltas can disrupt free-space connectivity. The continuous cost formulation is beneficial, but its calibration must reflect planner needs; in practice we find that (i) preserving dynamic range and mid-cost structure, and (ii) selecting an appropriate binarization threshold with modest inflation are decisive for navigation outcomes. By architecture: UNet is strongest at boundary fidelity and small objects (good for round/thin obstacles) but needs calibration to avoid conservative halos; ViT best preserves global layout and corridor continuity (robust to occlusions) yet can smooth very narrow gaps; Hybrid provides the best balance for mixed-scale outdoor scenes, sustaining connectivity at planner thresholds without sacrificing edge sharpness.

5.4 Ablation and Robustness Studies

5.4.1 Modality Ablation (H1): RGB vs RGBD

We hypothesize (H1) that providing depth information alongside RGB channels will lead to more accurate and reliable costmap generation compared to RGB-only models.

AUPRC Analysis. To provide a comprehensive evaluation across all decision thresholds, we compare the Area Under the Precision-Recall Curve (AUPRC) for both modality variants (Table 7). The AUPRC results strongly support H1: the addition of depth provides a substantial performance uplift on both datasets, with a 57.7% improvement on NYU and a 48.3% improvement on KITTI.

Table 7: ViT modality ablation: AUPRC comparison. RGBD vs RGB-only across datasets.

Dataset	ViT RGB (AUPRC)	ViT RGBD (AUPRC)	Δ AUPRC	% Improvement
NYU	0.4713	0.7432	+0.2719	+57.7%
KITTI	0.3085	0.4576	+0.1491	+48.3%

Detailed single-point metrics at $\tau = 0.5$ are provided in Appendix 7 (Table 14). RGBD reduces MAE on NYU and KITTI and improves precision/IoU, while RGB sometimes shows higher recall/F1 at $\tau = 0.5$. This behavior reflects a precision–recall trade-off and different optimal thresholds, underscoring that AUPRC better captures overall performance and that RGBD produces safer, more confident costmaps. These results confirm H1: RGBD lowers MAE, boosts AUPRC, and resolves visual ambiguities, making it the preferred choice for safe navigation.

5.4.2 Cross-Domain Generalization

Table 8: UNet cross-domain evaluation on NYU. Threshold $\tau = 0.5$. TL = transfer learning, ZS = zero-shot.

Training Scenario	MAE $\downarrow$	IoU $\uparrow$	Precision	Recall	F1
NYU only	0.0139	0.9168	0.9526	0.9607	0.9566
KITTI $\rightarrow$ NYU (TL)	0.0124	0.9219	0.9544	0.9645	0.9593
KITTI $\rightarrow$ NYU (ZS)	0.1437	0.4150	0.6357	0.5488	0.5858

Table 9: UNet cross-domain evaluation on KITTI. Threshold $\tau = 0.5$. TL = transfer learning, ZS = zero-shot.

Training Scenario	MAE $\downarrow$	IoU $\uparrow$	Precision	Recall	F1
KITTI only	0.2083	0.4803	0.6148	0.6989	0.6384
NYU $\rightarrow$ KITTI (TL)	0.2015	0.4999	0.6313	0.6956	0.6514
NYU $\rightarrow$ KITTI (ZS)	0.1852	0.4283	0.6382	0.5181	0.5676

Fine-tuning a KITTI model on NYU recovers almost all in-domain performance (F1 0.959 vs 0.957) whereas zero-shot deployment collapses to 0.586 F1 despite similar precision (Table 8). Outdoors the benefit of adaptation is even larger: NYU $\rightarrow$ KITTI transfer lifts IoU/F1 from (0.480, 0.638) to (0.500, 0.651), while zero-shot transfer lags by over eight F1 points (Table 9). Cross-domain pretraining therefore remains useful, but lightweight fine-tuning is essential for preserving free-space calibration under shifts in texture, depth, and illumination.

Table 10: ViT cross-domain evaluation on NYU. Threshold $\tau = 0.5$. TL = transfer learning, ZS = zero-shot.

Training Scenario	MAE $\downarrow$	IoU $\uparrow$	Precision	Recall	F1
NYU only	0.0019	0.9939	0.9963	0.9976	0.9970
KITTI $\rightarrow$ NYU (TL)	0.0023	0.9958	0.9991	0.9967	0.9979
KITTI $\rightarrow$ NYU (ZS)	0.4364	0.1129	0.1723	0.2463	0.2024

Table 11: ViT cross-domain evaluation on KITTI. Threshold $\tau = 0.5$. TL = transfer learning, ZS = zero-shot.

Training Scenario	MAE $\downarrow$	IoU $\uparrow$	Precision	Recall	F1
KITTI only	0.1694	0.5397	0.6996	0.6238	0.6543
NYU $\rightarrow$ KITTI (TL)	0.1822	0.5392	0.6956	0.6315	0.6574
NYU $\rightarrow$ KITTI (ZS)	0.1679	0.5412	0.7128	0.6055	0.6492

ViT transfer results reveal strong asymmetry: KITTI $\rightarrow$ NYU zero-shot fails (IoU 0.113, -88.6%), requiring fine-tuning to fully recover performance (Table 10). In contrast, NYU $\rightarrow$ KITTI zero-shot succeeds with near-baseline IoU (0.541 vs 0.540) and fine-tuning adds minimal benefit (Table 11). This confirms indoor scenes with dense depth provide superior pre-training: the ViT’s global attention effectively exploits dense geometric information but fails when depth characteristics mismatch training distributions (sparse LiDAR vs dense RGB-D).

5.4.3 Objective Ablation (H3)

Table 12: UNet objective ablation. Threshold $\tau = 0.5$. Compares L1-only, L1+Dice, and L1+Dice+Boundary loss configurations.

Dataset	Loss Config	MAE ↓	IoU ↑	Precision	Recall	F1
NYU	L1 only	0.0104	0.9237	0.9525	0.9683	0.9603
NYU	L1 + Dice	0.0261	0.9229	0.9392	0.9817	0.9599
NYU	L1 + Dice + Boundary	0.0204	0.9200	0.9584	0.9584	0.9583
KITTI	L1 only	0.1901	0.4135	0.6324	0.5580	0.5682
KITTI	L1 + Dice	0.2224	0.4411	0.6022	0.6549	0.6058
KITTI	L1 + Dice + Boundary	0.2216	0.4886	0.5921	0.7406	0.6472

On NYU the three losses differ by less than 0.3 F1 points—L1-only delivers the best MAE (0.0104) and essentially matches L1+Dice. On KITTI, however, Dice improves recall (0.655 vs 0.558) at the expense of higher MAE, and the boundary term is required to recover precision, raising IoU from 0.413 to 0.489 and F1 from 0.568 to 0.647. These results validate H3: overlap/boundary terms are critical for cluttered outdoor scenes, while the smoother indoor labels can be learned with a simple L1 objective.

For ViT (Appendix 7, Table 16): Across NYU, all losses achieve near-perfect F1, with L1-only giving the best MAE and precision, indicating minimal regularization is needed indoors. On KITTI, adding Dice and Boundary offers small but consistent gains in IoU and precision, improving thin-obstacle delineation despite similar F1 scores. Overall, segmentation-aware losses help sharpen boundaries when geometric cues dominate.

5.4.4 Robustness to Photometric and Geometric Perturbations

Table 13: Robustness evaluation. Clean metrics and perturbation deltas (relative to clean). Threshold $\tau = 0.5$.

Model	Dataset	Perturbation	Clean F1	Clean IoU	$\Delta F1$	ΔIoU
Classical Baseline	KITTI	Clean	0.6568	0.4987	–	–
Classical Baseline	KITTI	Brightness Darker	–	–	+0.0000	+0.0000
Classical Baseline	KITTI	Brightness Brighter	–	–	+0.0000	+0.0000
Classical Baseline	KITTI	Blur Sigma1	–	–	+0.0000	+0.0000
Classical Baseline	KITTI	Blur Sigma2	–	–	+0.0000	+0.0000
Classical Baseline	KITTI	Perspective Shift	–	–	+0.0244	+0.0286
Classical Baseline	NYU	Clean	0.9934	0.9869	–	–
Classical Baseline	NYU	Brightness Darker	–	–	+0.0000	+0.0000
Classical Baseline	NYU	Brightness Brighter	–	–	+0.0000	+0.0000
Classical Baseline	NYU	Blur Sigma1	–	–	+0.0000	+0.0000
Classical Baseline	NYU	Blur Sigma2	–	–	+0.0000	+0.0000
Classical Baseline	NYU	Perspective Shift	–	–	-0.1499	-0.2575
Learned	KITTI	Clean	0.6390	0.4747	–	–
Learned	KITTI	Brightness Darker	–	–	-0.5847	-0.4465
Learned	KITTI	Brightness Brighter	–	–	-0.5847	-0.4465
Learned	KITTI	Blur Sigma1	–	–	-0.5851	-0.4467
Learned	KITTI	Blur Sigma2	–	–	-0.5852	-0.4467
Learned	KITTI	Perspective Shift	–	–	-0.5652	-0.4359
Learned	NYU	Clean	0.9569	0.9175	–	–
Learned	NYU	Brightness Darker	–	–	-0.2311	-0.3479
Learned	NYU	Brightness Brighter	–	–	-0.2311	-0.3479
Learned	NYU	Blur Sigma1	–	–	-0.2312	-0.3479
Learned	NYU	Blur Sigma2	–	–	-0.2312	-0.3480
Learned	NYU	Perspective Shift	–	–	-0.2288	-0.3450

The classical depth baseline is almost invariant to brightness and blur because it operates directly on normalized depth, though perspective jitter reduces NYU F1 by 0.15. The learned RGBD UNet is markedly more brittle: darkening or blurring NYU inputs removes ≈ 0.23 F1/0.35 IoU, and the same perturbations on KITTI remove ≈ 0.58 F1/0.45 IoU. Perspective jitter also erodes KITTI F1 by

0.57. These observations motivate the integration of photometric augmentation and/or depth-only auxiliary heads if the learned costmaps are to match the heuristic pipeline’s stability in the wild.

5.5 Qualitative Visualizations

Representative panels (RGB, depth, ground-truth, predicted costmap) are shown in Figure 3; our visualization utility renders the modalities side-by-side to highlight residual errors. Additional sequences, modality ablations, and KITTI panels are collected in Appendix 7 to keep the main narrative concise.

5.5.1 Analysis of Costmaps

NYU ground-truth costmaps present a smooth traversability surface: costs increase toward cluttered, near-range regions and decrease in open floor areas. This reflects our depth-derived label heuristic (distance-transform with inflation), which favors continuous gradients over binary occupancy. RGBD predictions recover the global gradient and relative ordering, while RGB-only predictions largely preserve free vs obstacle ordering but flatten intermediate costs. This subtle loss degrades planner success (0.794–0.885 vs ≥ 0.969 for RGBD/labels) despite similar F1, indicating the planner is sensitive to cost calibration in connectivity-critical zones. The absence of sharp high-cost blobs suggests the heuristic emphasizes global layout rather than per-object peaks; boundary-aware or semantic priors could sharpen localized obstacles.

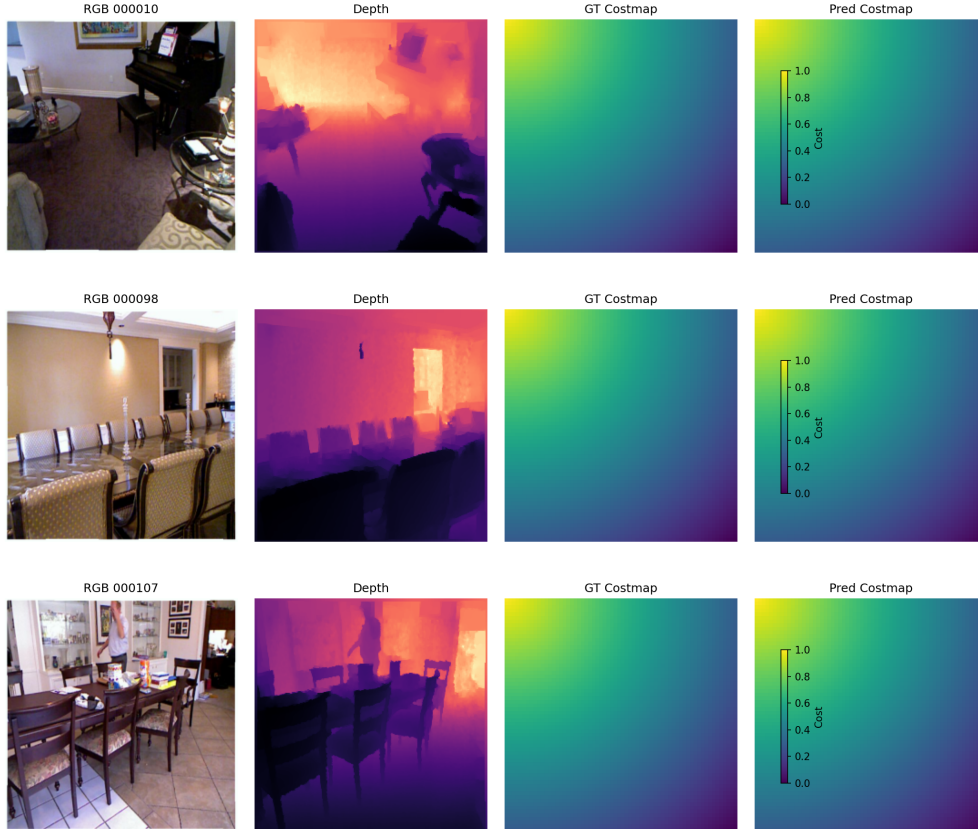


Figure 3: NYU RGBD predictions (baseline). Predictions closely match label gradients with minor residuals, yielding planner success ≥ 0.969 across thresholds.

6 Conclusion and Limitations

We presented a unified pipeline for learning continuous costmaps from RGBD and evaluating them with planners. On NYU, learned costmaps match labels both in pixel metrics and planning out-

comes; on KITTI, predictions are competitive but planner success depends on thresholding and calibration. ViT leads NYU IoU/F1, while Hybrid yields the highest KITTI IoU; RGB-only ablations retain good pixel metrics but reduce planner success, evidencing the importance of mid-cost calibration for connectivity. Architectural takeaways: use UNet when fine boundary detail and small/round obstacle fidelity are paramount (with planner-aware calibration), ViT when global context and occlusion reasoning dominate (indoors/structured layouts), and Hybrid for balanced performance in mixed-scale, outdoor domains where both crisp edges and long-range context matter.

Limitations include: (1) the depth-derived label heuristic emphasizes global gradients and may miss thin obstacles; (2) KITTI exhibits threshold sensitivity and residual miscalibration; (3) collision counts are implicit via planner failures rather than reported explicitly. Recommendations for deployment include data-driven threshold tuning with inflation radius 2, light morphological post-processing, and maintaining inference guardrails (distribution logging and degeneracy checks) to ensure usable predictions.

Future work entails exploring a start/goal-aware training curricula that bias supervision toward the sampling distribution used by planners, boundary-refinement heads or auxiliary contour losses to tighten KITTI calibration without inflating conservative halos, and validation on hardware by coupling the learned costmaps to a low-speed rover or indoor mobile base. These extensions remain compatible with the existing pipeline and would directly test transfer to real robots.

7 Team Contributions

Member	Technical contributions
Rut Santana	Data pipeline; processed pairs; configs; planner eval (A*/RRT*); modality ablation; report assembly; reproducibility orchestration.
Ibrahim Alshayeb	Hybrid model design/implementation; NYU/KITTI training; profiling; loss ablations.
Vineet Kulkarni	Classical depth baseline; UNet training (NYU/KITTI + transfer); robustness tests; baseline vs learned comparison.
Meera Ranjan	ViT implementation and training; PR curves/calibration; cross-domain experiments; qualitative panels.

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Appendix

Modality Ablation: Detailed Metrics

Table 14: ViT modality ablation at $\tau = 0.5$. RGBD vs RGB-only across NYU and KITTI datasets.

Dataset	Metric	ViT RGBD	ViT RGB	Δ (RGBD - RGB)	Note
NYU	MAE $\downarrow$	0.0019	0.0032	-0.0013	RGBD: 41% lower
	IoU $\uparrow$	0.9939	0.9957	-0.0018	RGB marginally better
	Precision	0.9963	0.9971	-0.0008	RGB marginally better
	Recall	0.9976	0.9985	-0.0009	RGB marginally better
	F1	0.9970	0.9978	-0.0008	RGB marginally better
KITTI	MAE $\downarrow$	0.1694	0.1988	-0.0294	RGBD: 14.8% lower
	IoU $\uparrow$	0.5397	0.5204	+0.0193	RGBD: 3.7% higher
	Precision	0.6996	0.6344	+0.0652	RGBD: 10.3% higher
	Recall	0.6238	0.6951	-0.0713	RGB: 11.4% higher
	F1	0.6543	0.6584	-0.0041	Nearly identical

Precision-Recall Curves

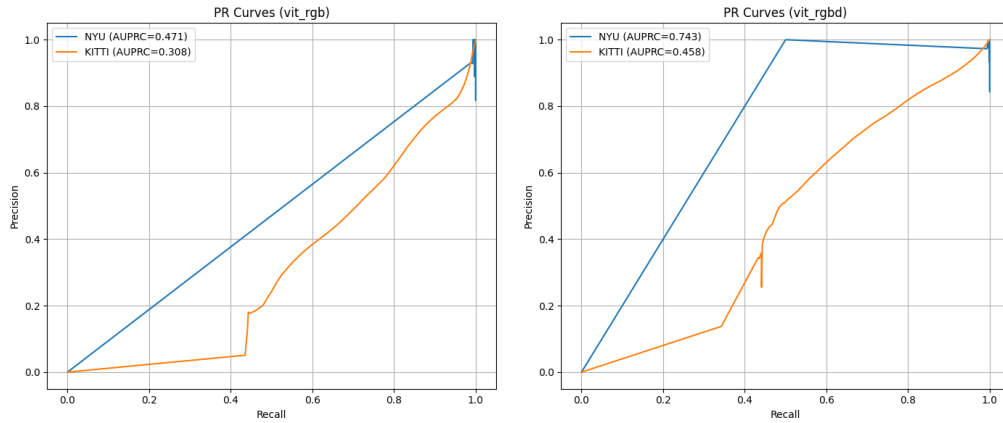


Figure 4: Precision-Recall curves for ViT RGB (left, AUPRC=0.471/0.309 NYU/KITTI) and ViT RGBD (right, AUPRC=0.743/0.458 NYU/KITTI). RGBD maintains higher precision at equivalent recall levels across both datasets, with particularly strong gains on NYU (+57.7%).

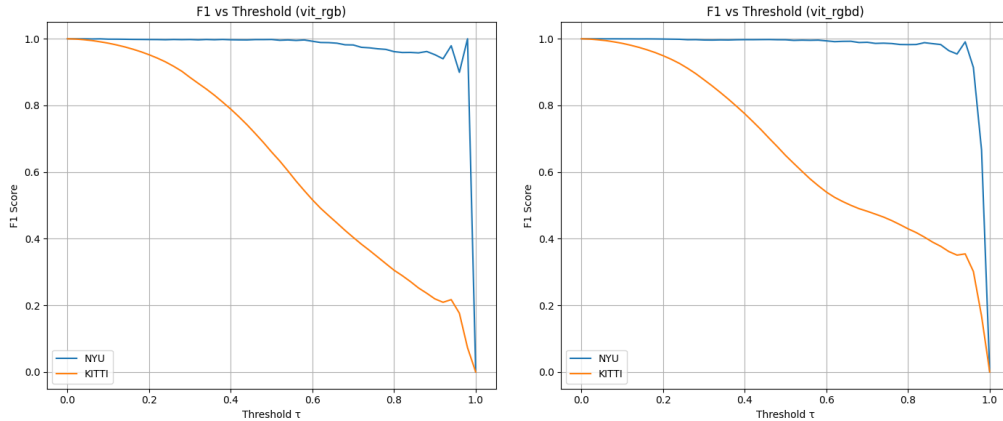


Figure 5: F1-score vs. Threshold for ViT RGB (left) and ViT RGBD (right). Both models show peak F1 at lower thresholds ($\tau \approx 0.2-0.4$), particularly on KITTI, justifying safety-oriented threshold selection for deployment.

Objective Ablation Table

Table 15: UNet objective ablation. Threshold $\tau = 0.5$. Compares L1-only, L1+Dice, and L1+Dice+Boundary loss configurations.

Dataset	Loss Config	MAE $\downarrow$	IoU $\uparrow$	Precision	Recall	F1
NYU	L1 only	0.0104	0.9237	0.9525	0.9683	0.9603
NYU	L1 + Dice	0.0261	0.9229	0.9392	0.9817	0.9599
NYU	L1 + Dice + Boundary	0.0204	0.9200	0.9584	0.9584	0.9583
KITTI	L1 only	0.1901	0.4135	0.6324	0.5580	0.5682
KITTI	L1 + Dice	0.2224	0.4411	0.6022	0.6549	0.6058
KITTI	L1 + Dice + Boundary	0.2216	0.4886	0.5921	0.7406	0.6472

Table 16: ViT objective ablation. Threshold $\tau = 0.5$. Compares L1-only, L1+Dice, and L1+Dice+Boundary loss configurations.

Dataset	Loss Config	MAE $\downarrow$	IoU $\uparrow$	Prec.	Recall	F1	Notes
KITTI	L1	0.1925	0.4877	0.5380	0.8522	0.6540	High recall, loose predictions
KITTI	L1 + Dice	0.1709	0.5366	0.6958	0.6249	0.6530	Best IoU & precision
KITTI	L1 + Dice + Boundary	0.1693	0.5168	0.6617	0.6611	0.6549	Best MAE & F1, sharp edges

Robustness Table

Table 17: Robustness evaluation. Clean metrics and perturbation deltas (relative to clean). Threshold $\tau = 0.5$.

Model	Dataset	Perturbation	Clean F1	Clean IoU	$\Delta F1$	ΔIoU
Classical Baseline	KITTI	Clean	0.6568	0.4987	–	–
Classical Baseline	KITTI	Brightness Darker	–	–	+0.0000	+0.0000
Classical Baseline	KITTI	Brightness Brighter	–	–	+0.0000	+0.0000
Classical Baseline	KITTI	Blur Sigma1	–	–	+0.0000	+0.0000
Classical Baseline	KITTI	Blur Sigma2	–	–	+0.0000	+0.0000
Classical Baseline	KITTI	Perspective Shift	–	–	+0.0244	+0.0286
Classical Baseline	NYU	Clean	0.9934	0.9869	–	–
Classical Baseline	NYU	Brightness Darker	–	–	+0.0000	+0.0000
Classical Baseline	NYU	Brightness Brighter	–	–	+0.0000	+0.0000
Classical Baseline	NYU	Blur Sigma1	–	–	+0.0000	+0.0000
Classical Baseline	NYU	Blur Sigma2	–	–	+0.0000	+0.0000
Classical Baseline	NYU	Perspective Shift	–	–	-0.1499	-0.2575
Learned	KITTI	Clean	0.6390	0.4747	–	–
Learned	KITTI	Brightness Darker	–	–	-0.5847	-0.4465
Learned	KITTI	Brightness Brighter	–	–	-0.5847	-0.4465
Learned	KITTI	Blur Sigma1	–	–	-0.5851	-0.4467
Learned	KITTI	Blur Sigma2	–	–	-0.5852	-0.4467
Learned	KITTI	Perspective Shift	–	–	-0.5652	-0.4359
Learned	NYU	Clean	0.9569	0.9175	–	–
Learned	NYU	Brightness Darker	–	–	-0.2311	-0.3479
Learned	NYU	Brightness Brighter	–	–	-0.2311	-0.3479
Learned	NYU	Blur Sigma1	–	–	-0.2312	-0.3479
Learned	NYU	Blur Sigma2	–	–	-0.2312	-0.3480
Learned	NYU	Perspective Shift	–	–	-0.2288	-0.3450

Planner Tables

Table 18: NYU planner-in-the-loop results across thresholds for the UNet RGB (RGB-only) model (source=predictions).

thr	succ(A*)	succ(RRT*)	t(A*) [ms]	t(RRT*) [ms]	len(A*)	len(RRT*)
0.4	0.794	0.794	1.099	40.396	34.702	41.121
0.5	0.809	0.809	0.874	44.351	27.927	34.212
0.6	0.885	0.885	1.185	35.273	32.800	41.069

Table 19: NYU planner-in-the-loop results across thresholds for the ViT RGB model (source=predictions).

thr	succ(A*)	succ(RRT*)	t(A*) [ms]	t(RRT*) [ms]	len(A*)	len(RRT*)
0.4	0.794	0.794	1.569	76.382	29.716	36.171
0.5	0.817	0.817	2.030	125.716	30.812	37.850
0.6	0.916	0.916	2.476	65.251	33.626	42.227

Table 20: NYU planner-in-the-loop results across thresholds for the ViT RGBD model (source=predictions).

thr	succ(A*)	succ(RRT*)	t(A*) [ms]	t(RRT*) [ms]	len(A*)	len(RRT*)
0.4	0.817	0.817	1.781	95.156	32.718	38.846
0.5	0.885	0.885	2.146	78.449	31.463	38.589
0.6	0.924	0.924	2.263	74.647	32.707	40.868

Table 21: KITTI planner-in-the-loop results across thresholds for the UNet RGB (RGB-only) model (source=predictions).

thr	succ(A*)	succ(RRT*)	t(A*) [ms]	t(RRT*) [ms]	len(A*)	len(RRT*)
0.4	0.497	0.478	0.755	45.232	30.374	33.743
0.5	0.587	0.573	0.952	42.095	33.191	37.497
0.6	0.603	0.594	1.338	50.468	37.986	43.848

Table 22: KITTI planner-in-the-loop results across thresholds for the UNet RGBD model (source=predictions).

thr	succ(A*)	succ(RRT*)	t(A*) [ms]	t(RRT*) [ms]	len(A*)	len(RRT*)
0.4	0.483	0.471	0.757	44.647	25.777	29.882
0.5	0.584	0.573	0.865	60.031	28.598	32.982
0.6	0.630	0.630	1.305	70.739	30.831	37.261

Table 23: KITTI planner-in-the-loop results across thresholds for the UNet RGBD retrain full model (source=predictions).

thr	succ(A*)	succ(RRT*)	t(A*) [ms]	t(RRT*) [ms]	len(A*)	len(RRT*)
0.4	0.543	0.540	0.684	45.826	28.151	32.378
0.5	0.644	0.642	1.159	50.768	34.488	41.477
0.6	0.794	0.785	1.277	48.624	35.649	43.176

Table 24: KITTI planner-in-the-loop results across thresholds for the ViT RGB model (source=predictions).

thr	succ(A*)	succ(RRT*)	t(A*) [ms]	t(RRT*) [ms]	len(A*)	len(RRT*)
0.4	0.564	0.561	0.910	62.572	28.864	32.342
0.5	0.693	0.686	1.147	68.122	28.365	33.231
0.6	0.781	0.781	1.433	82.675	29.984	37.184

Table 25: KITTI planner-in-the-loop results across thresholds for the ViT RGBD model (source=predictions).

thr	succ(A*)	succ(RRT*)	t(A*) [ms]	t(RRT*) [ms]	len(A*)	len(RRT*)
0.4	0.739	0.739	1.195	75.310	28.611	33.531
0.5	0.861	0.861	1.526	64.040	30.764	37.669
0.6	0.887	0.887	1.737	60.053	32.700	40.050

Additional Qualitative Panels

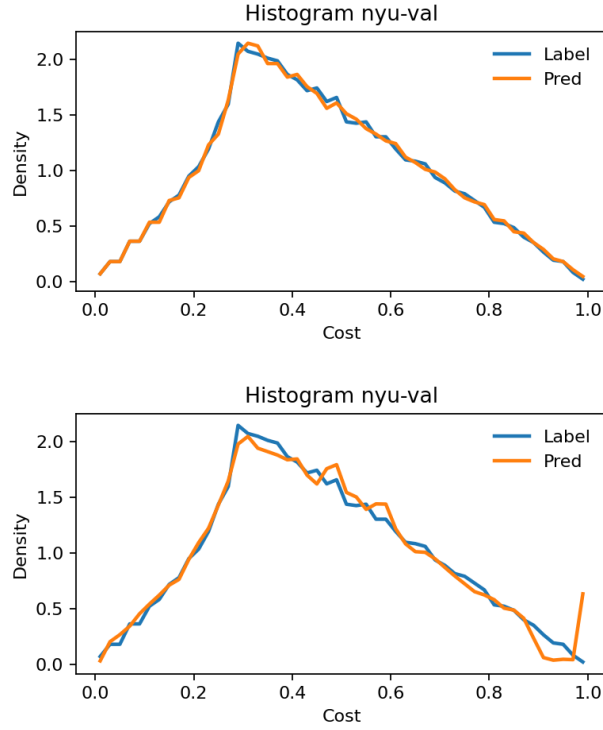


Figure 6: Cost distribution histograms (NYU val, first 50). Top: RGBD baseline closely matches label distribution ($KL_{L||P} \approx 9e-4$). Bottom: RGB-only deviates modestly ($KL_{L||P} \approx 2.26e-2$), consistent with planner success gap.

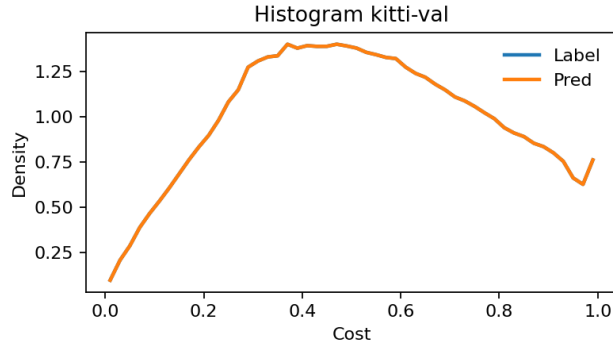


Figure 7: KITTI validation prediction cost distribution ($n = 433$). Shown is the *pre-fix* distribution that was saturated near cost 1.0, explaining zero planner success. After the inference fix (Section 5.3), the distribution preserves dynamic range ($KL_{L||P} = 0.090$, $KL_{P||L} = 0.056$) and supports A*/RRT* success ≈ 0.80 at $\tau = 0.5$ (full validation set).

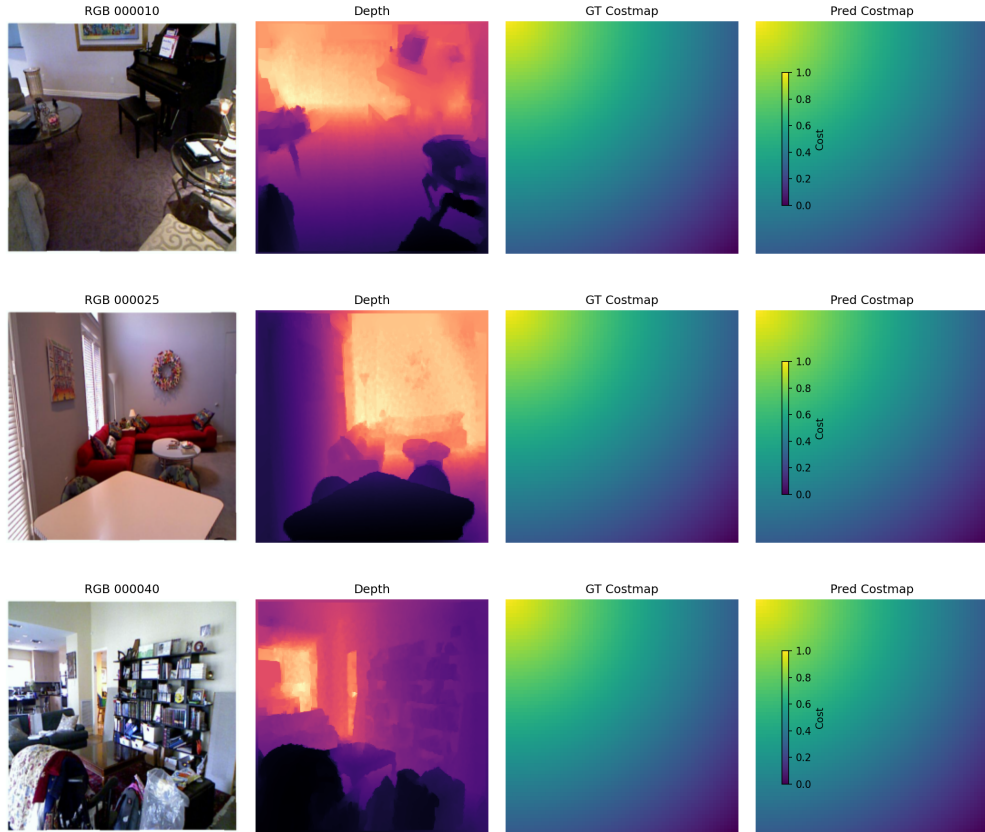


Figure 8: Representative NYU RGB, depth, ground-truth, and prediction panels rendered side-by-side for qualitative inspection.

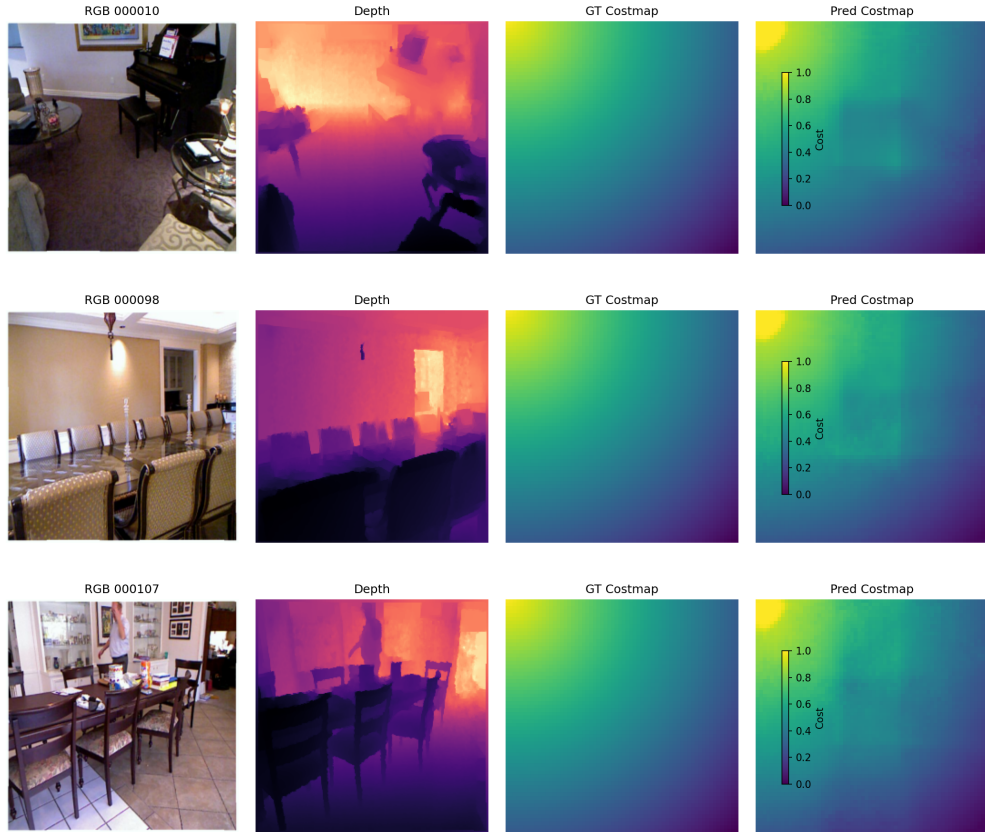


Figure 9: NYU RGB-only predictions (UNet). Coarse structure is preserved but intermediate costs are flattened, which reduces planner success compared to RGBD.

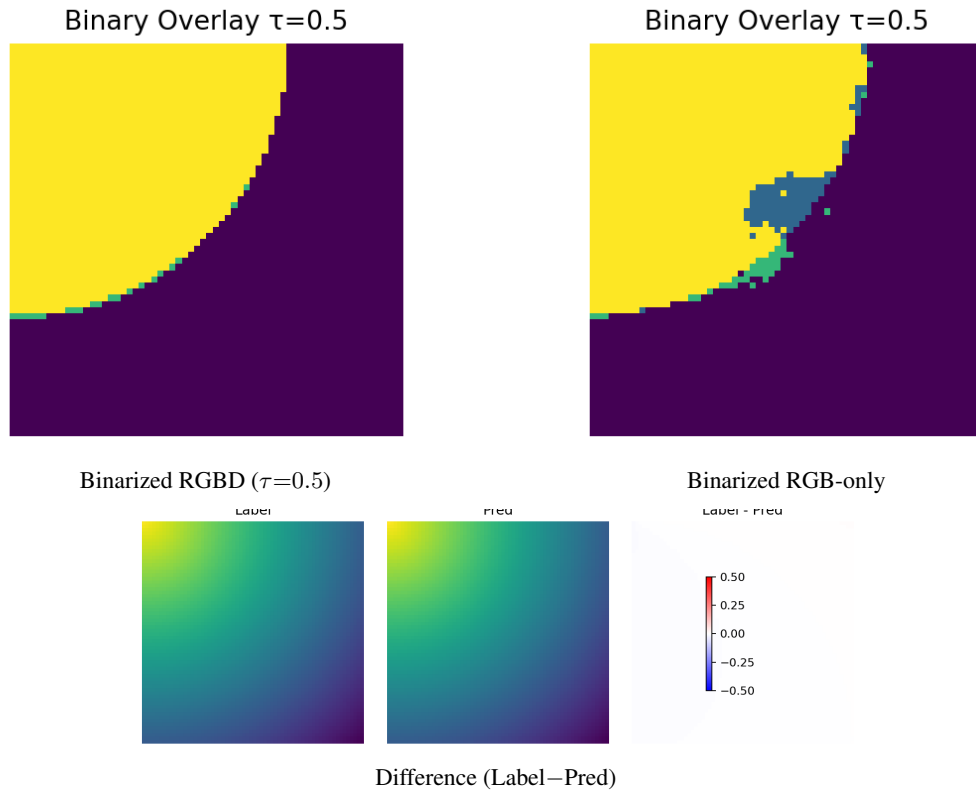


Figure 10: NYU appendix panels. RGBD predictions preserve mid-cost structure and produce clean free-space after binarization; the RGB-only model flattens mid-costs, which reduces connectivity and planner success compared to RGBD.

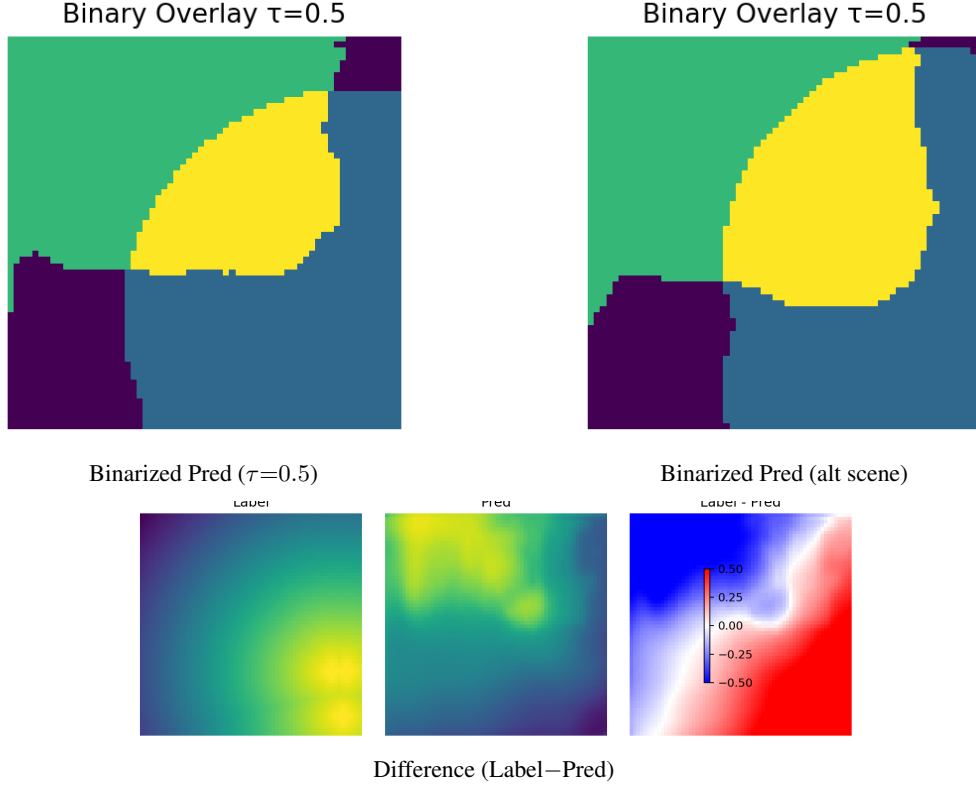


Figure 11: KITTI appendix panels (post-inference fix). Binarized predictions (inflation 2, $\tau \in [0.5, 0.6]$) show restored free-space connectivity; difference maps highlight residual false positives near start/goal pockets, aligning with ≈ 0.73 – 0.84 success (Tables 5–6).

Planner Path Overlays

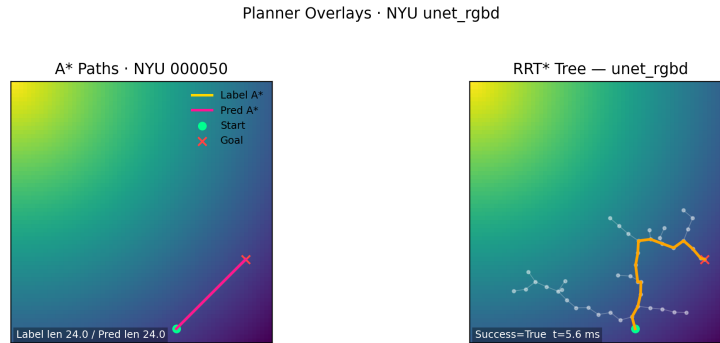


Figure 12: NYU planner overlays for the UNet RGBD model (scene 000050). Label and prediction A* paths overlap closely, while the RRT* tree confirms that mid-cost gradients remain intact indoors.

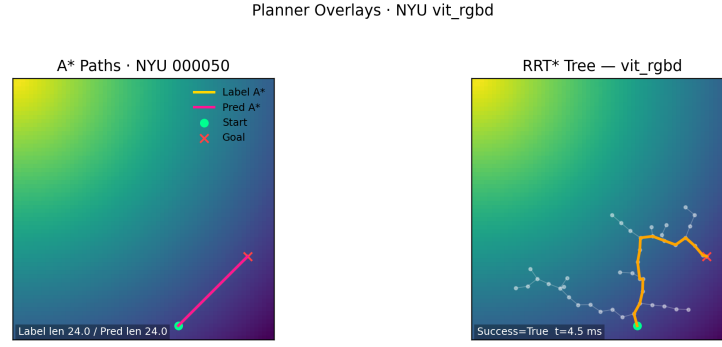


Figure 13: NYU planner overlays for the ViT RGBD model (scene 000050). The ViT maintains the corridor structure and produces a compact RRT* tree with minimal detours, illustrating its global context advantages.

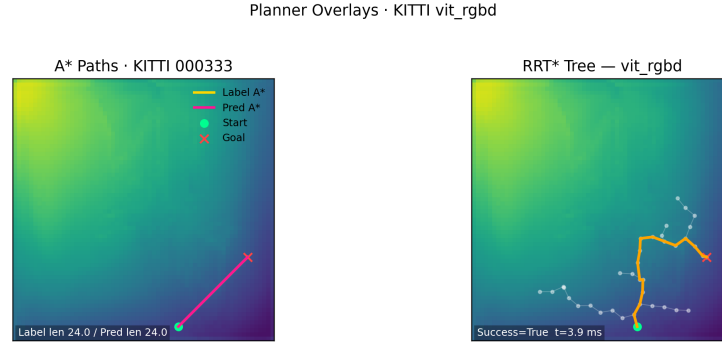


Figure 14: KITTI planner overlays for the ViT RGBD model (scene 000333). Slightly smoother A* trajectories and a wider RRT* tree highlight the remaining calibration gap relative to the UNet retrain in Figure 1.

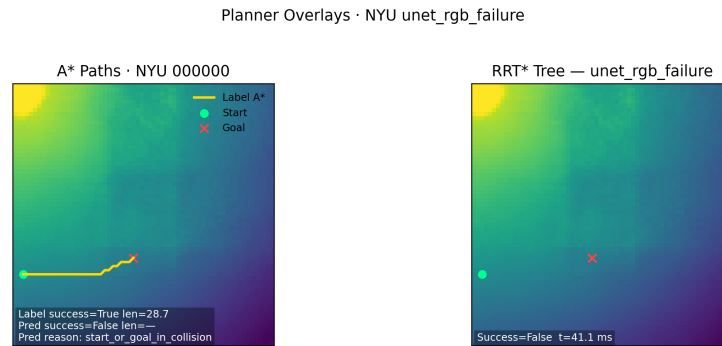


Figure 15: NYU failure-case overlay for the UNet RGB model (scene 000000). Labels provide a clean path, but the RGB-only prediction inflates the doorway and blocks A*. The RRT* tree terminates near the start, mirroring the logged planner failure.

Planner Overlays · NYU vit_rgbd_failure

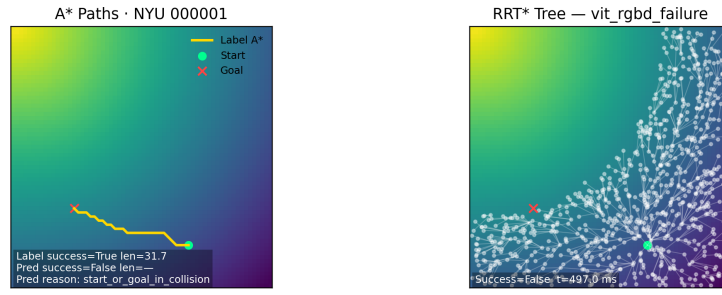


Figure 16: NYU failure-case overlay for the ViT RGBD model (scene 000001). The label path remains open, yet the ViT prediction raises a thin high-cost band that prevents both A* and RRT* from leaving the start, illustrating sensitivity to corridor pinch points.

Planner Overlays · KITTI vit_rgbd_failure

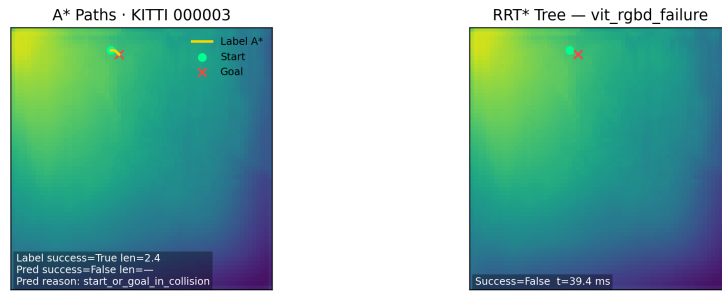


Figure 17: KITTI failure-case overlay for the ViT RGBD model (scene 000003). Label A* reaches the goal, but the predicted map introduces a high-cost island around the start so neither planner recovers, matching the per-scene failure logs.